

Experiment 4:

Implement Individual Conditional Expectation (ICE) plots and Accumulated Local Effects (ALE) plots to analyze and interpret the effect of input features on the predictions of a trained nonlinear machine-learning model.

Instructions:

1. Select tabular data.
2. Perform the required preprocessing for the selected data.
3. Split the data into training and testing.
4. Train a nonlinear machine-learning model, such as a Decision Tree, Random Forest, or Gradient Boosting model.
5. Evaluate the trained model on the test data using appropriate performance metrics.
6. Select one numerical feature for ICE analysis.
7. Determine an appropriate range for the feature, preferably from the observed data distribution.
8. Generate K equally spaced values for the selected range as a vector R.
9. Make a copy of test data.
10. Replace the entire selected feature column by one value from the R and test the model with this set.
11. Store the predictions corresponding to all values of R.
12. Repeat the procedure for all selected test samples.
13. Plot the selected feature values R on the x-axis and the corresponding predictions on the y-axis, with one curve for each individual sample.
14. Observe and interpret the differences among the individual ICE curves.

For ALE plots:

1. Select a numerical feature for ALE analysis.
2. Divide the observed range of the selected feature into suitable intervals or bins.
3. For the samples belonging to each interval, replace the selected feature value with the lower and upper boundaries of that interval.
4. Obtain model predictions at both boundaries and calculate the difference in predictions.
5. Calculate the average prediction difference for all samples belonging to each interval.
6. Accumulate the average local effects across the intervals.
7. Center the accumulated effects around zero to obtain the ALE values.
8. Plot the selected feature values or intervals on the x-axis and the corresponding ALE values on the y-axis.
9. Interpret how the selected feature affects the model prediction across different regions of its distribution.